Tender Ref ID: 999239 | Notice No: 23/MD/CE-Assembly/AGICL/2026 , Dated: 28.09.2026
Estimated Contract Value: Rs. 3,62,94,39,875/- | Bid Closing: 29/10/2026 03:00 PM
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VOLUME III · CONCRETE MIX DESIGN, REBAR & POST-TENSIONING

1. HIGH-PERFORMANCE STRUCTURAL CONCRETE:
? Grades: M50 for foundation raft and columns; M60 Self-Compacting Concrete (SCC) for flared shear walls and core.
? Cement: 53-Grade OPC adhering to IS 269:2015 with 25% Fly Ash / GGBS pozzolanic replacement.
? Aggregates: Graded crushed angular granite aggregates adhering to IS 383:2016.
? Superplasticizer: Polycarboxylate Ether (PCE) based superplasticizers adhering to IS 9103.

2. REINFORCEMENT & POST-TENSIONING SYSTEM:
? Rebar: Fe550D TMT bars adhering to IS 1786 with minimum elongation of 14.5%.
? PT Strands: Low-relaxation 7-wire strands conforming to IS 14268 Class 2 (Nominal Dia 15.2mm, UTS 1860 MPa).
? Grouting: Vacuum-assisted pressure grouting using non-shrink polymer grout adhering to EN 447.

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VOLUME III · MODULAR EXPRESS FORMWORK & SURFACE FINISHES

3. MODULAR FAIR-FINISH EXPRESS FORMWORK:
? Engineered System: High-capacity modular fair-finish express formwork (Doka / Peri / Meva system).
? Surface Benchmark: Architectural F4 fair-face finish with uniform chamfered edges and concealed tie-rod cones.
? Permissible Surface Deviation: Not exceeding 2mm over a 3.0-meter straight edge benchmark.

4. HIGH-STRENGTH STRUCTURAL STEEL SPIRE FABRICATION:
? Materials: Seamless hot-finished structural hollow sections in steel Grade S355J2H / S460NH conforming to EN 10210.
? Corrosion Protection: Multi-coat epoxy zinc-rich primer (75 microns) + micaceous iron oxide intermediate (100 microns) + aliphatic polyurethane topcoat (75 microns) with total dry film thickness (DFT) of not less than 250 microns.

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VOLUME III · TUNED MASS DAMPER & INCLINED PASSENGER ELEVATORS

5. TUNED MASS DAMPER (TMD) SPECIFICATION:
? Configuration: 35-Ton bidirectional pendulum mass damper installed at +210.0m level inside spire cavity.
? Tuning Frequency: Calibrated to primary lateral sway mode of 0.28 Hz.
? Performance Target: Top-level lateral acceleration under 10-year wind not exceeding 12 milli-g (ISO 6897 limit: 18 milli-g).

6. INCLINED PANORAMIC PASSENGER LIFTS:
? Speed: 2.5 m/s variable inclination rope traction system with full regenerative drive.
? Glass Enclosure: Laminated safety glass with UV solar reflective coating.